

EN.605.734 section 81 Syllabus

Human-in-the-Loop AI: Crowdsourcing, LLMs, and Evaluation

Course Information

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Human-in-the-Loop AI: Crowdsourcing, LLMs, and Evaluation

EN.605.734.81 (3.0 Credits)

AE Fall 2026 [AE Fall 2026]

Description

Human-in-the-Loop AI: Crowdsourcing, LLMs, and Evaluation explores how modern AI systems are built, improved, and audited through the combined strengths of people and machines. As large language models reshape the way organizations collect data, make decisions, and deploy intelligent systems, this course examines when human judgment still matters, when AI can automate complex tasks, and when hybrid human-AI workflows produce the best results. Students will study crowdsourcing, task and prompt design, data quality, inter-annotator agreement, evaluation methods, and bias in both human- and AI-generated outputs. Through hands-on assignments and applied projects, students will compare crowd workers, LLMs, and human-AI pipelines on realistic problems in classification, annotation, conversational systems, and responsible AI evaluation. The course emphasizes practical skills in designing reliable workflows, measuring quality, and identifying failure modes in intelligent systems. By the end of the course, students will be able to build and assess human-in-the-loop AI systems that are effective, scalable, and trustworthy.

Department: PE Computer Science

College: Engineering and Applied Science Programs for Professionals

Instructor Information:

Instructors



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Communication Policy:

*I prefer that students contact me via email. Please be sure to include course number in the subject line. I will make every effort to respond to your inquiry within **48 hours** or earlier. If an issue is urgent, please indicate "urgent" within the subject line of the email and I will respond as soon as is practical.*

Office Hours:

Please refer to Canvas for the Zoom link and time for weekly synchronous chat, which we call "Bits & Bytes". This is optional. We will have students from around the world and this is an asynchronous course, so we do not want to disadvantage anyone by introducing new content. If anything meaningful is discussed, recordings will be posted. If you don't see a recording, it just means we were chatting about non-course-relevant stuff.

Course Structure:

The course materials are divided into modules which can be accessed by clicking **Modules** on the course menu. A module will have several sections including the overview, content, readings, discussions, and assignments. You are encouraged to preview all sections of the module before starting. Most modules run for a period of seven (7) days. You should regularly check the **Calendar** and **Announcements** for assignment due dates.

Course Topics:

Intro to Social Computing--connecting people through technology

Human Computation and Agentic AI Systems

AI Assisted Coding

Task & Prompt Design for Human & AI Workers

Human Judgement, Expertise & Decision Making

Crowdsourcing in Practice

Platforms

Inter-annotator Agreement (IAA)

Crowdsourcing and Machine Learning

Building Human AI Systems

LLMs and Crowdsourcing

Bias and Framing in LLMs

Human in the Loop Governance

Final paper and project presentation

Course Goals:

1. Analyze the principles of human computation, human-AI collaboration, and Human-in-the-Loop (HITL) system design, explaining how humans and AI contribute complementary computational capabilities.
2. Design human-AI systems that appropriately allocate tasks among humans, AI models, and hybrid workflows while incorporating effective human oversight, quality assurance, and governance mechanisms.
3. Evaluate AI systems using empirical methods including human evaluation, crowdsourcing, prompt evaluation, benchmarking, and experimental design to assess system performance, reliability, and robustness.
4. Assess the ethical, societal, and organizational implications of AI deployment, including bias, transparency, explainability, trust, governance, and responsible AI practices.

5. Apply modern AI development techniques (AI-assisted coding, prompt engineering, workflow optimization, and agentic AI design) to create reliable and effective AI-enabled applications.
6. Critically compare the strengths and limitations of humans, crowds, LLMs, and hybrid human-AI systems to determine appropriate computational approaches for diverse real-world problems.
7. Develop and communicate evidence-based recommendations for the design, evaluation, and governance of Human-in-the-Loop AI systems through technical analysis, experimentation, and written reports.

Course Learning Outcomes (CLOs):



No Course Learning Outcomes are available for this course.

Required Text and Other Materials

Textbooks:

No textbook is required.

You will have to purchase your own AWS credits for crowdsourcing assignments.

Sheridan Libraries:

The [Johns Hopkins Sheridan Libraries](#) provides a curated list of resources for Engineering students in their [Engineering for Professionals Guide](#).

For research help, refer to the [Research How-To Guides](#) or [schedule a research consultation](#).

Some, though not all, electronic versions of textbooks may be available through the Sheridan Libraries. **There is no guarantee that an electronic textbook is available.** Please email asklib@jhu.edu if you have any questions about searching for available eBooks.

Technical Requirements:

You should refer to [General Technical Requirements](#) for guidance on system requirements. Access support resources from the **Help** menu if you encounter any technical issues.

Evaluation and Grading

Student Coursework Requirements:

One page reflections 20%

Assignments 50%

Discussions 10%

Grading Policy:

EP uses a +/- grading system (see "Grading System", *Graduate Programs* catalog, p. 10).

Score Range	Letter Grade
100-97	= A+
<97-93	= A
<93-90	= A–
<90-87	= B+
<87-83	= B
<83-80	= B–
<80-77	= C+
<77-73	= C
<73-70	= C–
<70-67	= D+
<67-63	= D
<63	= F

Policies

Student Generative AI (GenAI) Use:

Green: Liberal GenAI use is permitted

Because developing proficiency with GenAI tools is a core learning outcome, students are expected to engage with these tools throughout the course. Additional guidance is provided below:

- **GenAI may be used throughout the workflow.** However, AI is not to be used to replace critical thinking. Follow specific guidelines for each assignment.
- **Disclosure of GenAI use is required.** When AI is used, include the following as an appendix to the assignment submission:
 1. A Reference using APA interview format. Example: *ChatGPT-o5. (Year, Month Day). Purpose of Interaction. (Student_Name, Interviewer)*
 2. The prompt(s) used and the responses provided (include the conversation). Follow specific guidelines for each assignment. A link to the conversation meets this requirement.
- **Student responsibility:** Students remain fully responsible for verifying accuracy, identifying bias, and ensuring the quality of final submissions.
 1. Assignments are designed to uncover the strengths and weaknesses of AI when used in support of Systems Engineering. To accomplish this objective, they incorporate both AI research and student analysis/assessment.
 2. Students are responsible for clearly separating AI-generated content from students' original thought/critical thinking in their submissions.

Unauthorized use of GenAI tools constitutes a violation of academic integrity policies. If you have questions about permitted GenAI use, please contact us. EP provides student guidance on responsible GenAI: (<https://ep.jhu.edu/student-services/academic-services/academic-policies/student-guidance-on-responsible-generative-ai-use/>).

Additional Resources:

Personal Wellbeing

JHU's [Student Health and Well-Being \(SHWB\)](#) has many resources available to support EP student wellbeing, including [mental health](#). Many students struggle with stress or a variety of mental health concerns. Life happens — please do not wait to seek assistance if you need it.

The Whiting School of Engineering also has a dedicated office for Student Support and Advocacy. [Engineering Student Support & Advocacy \(ESSA\)](#) helps students navigate non-Academic issues including mental or physical health, interpersonal issues, conflict with advisors, financial concerns, time management, leaves of absence, being victimized, and family emergencies.

As a student, you have access to [TimelyCare](#) TalkNow, which offers on-demand mental health support as well as up to 12 free counseling appointments with the provider of your choice. Psychiatric care is also available through TimelyCare for routine medication management (no stimulants or other controlled substances).

Telephone support from licensed clinicians is available 24/7/365 by calling [Johns Hopkins University Behavioral Health Crisis Support Team \(BHCST\)](#) at **410-516-9355 (WELL)**.

Tutoring Website

Johns Hopkins Engineering for Professionals offers a tutoring connection network that allows students to connect with other Johns Hopkins Engineering students or alumni for tutoring services. This service allows students to search a list of courses to “Find a Tutor” or complete a profile to “Become a Tutor.” More information about this service can be found on the [tutoring website](#).

Student Academic Success Office

If you are struggling academically, experience extenuating circumstances, or need additional academic support during the semester, the [EP Student Academic Success Office \(SASO\)](#) is here to help. Please email ep-studentsuccess@jhu.edu to be connected with a Student Academic Success Coordinator.

Privacy Policies:

To learn more about how to protect your data and privacy, visit [Instructure's privacy policy](#) (Canvas) and [JHU's privacy policy](#).

Canvas Accessibility:

Online courses are taught in the Canvas learning management system. To learn more about how Canvas is designed to be accessible, visit [Canvas's accessibility standards](#)

Academic Policies:



Deadlines for Adding, Dropping, and Withdrawing from Courses

Students may add a course up to one week after the start of the term for that particular course. Students may drop courses according to the drop deadlines outlined in the [EP academic calendar](#). Between the 6th week of the class and prior to the final withdrawal deadline, a student may withdraw from a course with a W on their academic record. A record of the course will remain on the academic record with a W appearing in the grade column to indicate that the student registered and withdrew from the course.



Academic Misconduct Policy

All students are required to read, know, and comply with the Johns Hopkins University Krieger School of Arts and Sciences (KSAS) / Whiting School of Engineering (WSE) [Procedures for Handling Allegations of Misconduct](#) by Full-Time and Part-Time Graduate Students.

This policy prohibits academic misconduct, including but not limited to the following: cheating or facilitating cheating; plagiarism; reuse of assignments; unauthorized collaboration; alteration of graded assignments; and unfair competition. Course materials (old assignments, texts, or examinations, etc.) should not be shared unless authorized by the course instructor. Any questions related to this policy should be directed to EP's academic integrity officer at ep-academic-integrity@jhu.edu.



Students with Disabilities – Accommodations and Accessibility

Johns Hopkins University values diversity and inclusion. We are committed to providing welcoming, equitable, and accessible educational experiences for all students. Our courses are designed with a proactive approach to accessibility to minimize the need for disability disclosure and accommodation requests, but we recognize that you may need additional support. Students with disabilities (including those with psychological conditions, medical conditions, and temporary disabilities) can request accommodations for this course by providing an Accommodation Letter issued by [Student Disability Services \(SDS\)](#). Please request accommodations for this course as early as possible to provide time for effective communication and arrangements.

For further information or to start the process of requesting accommodations, please contact EP Student Disability Services at ep-disability-svcs@jhu.edu.



Student Conduct Code

The fundamental purpose of the JHU regulation of student conduct is to promote and to protect the health, safety, welfare, property, and rights of all members of the University community as well as to promote the orderly operation of the University and to safeguard its property and facilities. As members of the University community, students accept certain responsibilities which support the educational mission and create an environment in which all students are afforded the same opportunity to succeed academically.

For a full description of the code please visit the [Student Conduct Code website](#).



Classroom Climate

JHU is committed to creating a classroom environment that values the diversity of experiences and perspectives that all students bring. Everyone has the right to be treated with dignity and respect. Fostering an inclusive climate is important. Research and experience show that students who interact with peers who are different from themselves learn new things and experience tangible educational outcomes. At no time in this learning process should someone be singled out or treated unequally on the basis of any seen or unseen part of their identity.

If you have concerns in this course about harassment, discrimination, or any unequal treatment, or if you seek accommodations or resources, please reach out to the course instructor directly. Reporting will never impact your course grade. You may also share concerns with your program chair, the Assistant Dean for Diversity and Inclusion, or the [Office of Institutional Equity](#). In handling reports, people will protect your privacy as much as possible, but faculty and staff are required to officially report information for some cases (e.g. sexual harassment).



Course Auditing

When a student enrolls in an EP course with "audit" status, the student must reach an understanding with the instructor as to what is required to earn the "audit." If the student does not meet those expectations, the instructor must notify the EP Registration Team (EP-Registration@exchange.johnshopkins.edu) in order for the student to be retroactively dropped or withdrawn from the course (depending on when the "audit" was requested and in accordance with EP registration deadlines). All lecture content will remain accessible to auditing students, but access to all other course material is left to the discretion of the instructor.